

Unit 2 Guided Notes — Scientific Notation

Big or tiny number? Pack it into a $\times 10^n$.

Standards: NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

1 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \leq a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45,000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent -3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	_____	$45,000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	_____	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE A Scientific Notation Conversions

Remember: $1 \leq a < 10$. Positive exp = BIG · Negative exp = SMALL.

1

45,000,000 → sci. notation

2

0.00032 → sci. notation

3

6.02×10^8 → standard

4

9.1×10^{-5} → standard

5

780,000 → sci. notation

7

 $3.45 \times 10^4 \rightarrow$ standard

9

5,280,000,000 → sci. notation

11

0.000000904 → sci. notation

13

63,400,000,000 → sci. notation

6

0.0000067 → sci. notation

8

 $1.2 \times 10^{-3} \rightarrow$ standard

10

 $2.5 \times 10^{-2} \rightarrow$ standard

12

 $7.15 \times 10^6 \rightarrow$ standard

14

 $8.8 \times 10^{-6} \rightarrow$ standard

Multiplying Scientific Notation — 3-Step Process

1. _____ the coefficients (the a-values)
2. _____ the exponents
3. Adjust if coefficient is NOT between _____ and _____

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ **Answer:** 6×10^9

Dividing Scientific Notation — 3-Step Process

1. _____ the coefficients
2. _____ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ ADJUSTMENT RULES:

- If coefficient ≥ 10 : Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1 : Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^5

Adjust ($20 \geq 10$): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3 .

Small number $\rightarrow$ NEGATIVE exponent. Correct: 4.5×10^{-3} . Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE B Operations with Scientific Notation

Multiply/divide coefficients · add/subtract exponents · adjust so $1 \leq a < 10$.

1

$(3 \times 10^4)(2 \times 10^5)$

2

$(8 \times 10^7) \div (4 \times 10^3)$

3

$(5 \times 10^3)(7 \times 10^6)$

4

$(9.6 \times 10^8) \div (3.2 \times 10^5)$

5

$(6 \times 10^5)(5 \times 10^4)$

6

$(1.5 \times 10^{-3})(4 \times 10^7)$

7

$(7.2 \times 10^9) \div (1.8 \times 10^{-2})$

8

$(2.5 \times 10^6)(8 \times 10^{-3})$

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE C Word Problems

Identify the operation, write your setup, then adjust so $1 \leq a < 10$.

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds for light to reach Earth?

Answer: _____

2. A bacteria cell is about 2×10^{-6} m long; a virus about 5×10^{-8} m. How many times longer is the bacteria?

Answer: _____

3. The US debt is about 3.4×10^{13} dollars; population about 3.4×10^8 . What is the debt per person?

Answer: _____

4. A grain of sand has mass 6.5×10^{-4} g. A beach has 8×10^{18} grains. Total mass in grams?

Answer: _____

5. A body has 3.7×10^{13} cells, each 1×10^{-5} m wide. Lined up, how long in meters?

Answer: _____

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12} . How many seconds?

Answer: _____

7. A hard drive holds 2×10^{12} bytes. Each photo is 4×10^6 bytes. How many photos fit?

Answer: _____

8. Light travels 3×10^8 m/s. How far in 6×10^3 seconds?

Answer: _____

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Scientific Notation Conversions

1

7,200,000 → sci. notation

3

$3.05 \times 10^7 \rightarrow$ standard

5

91,000 → sci. notation

7

$4.44 \times 10^6 \rightarrow$ standard

2

0.00045 → sci. notation

4

$6.1 \times 10^{-5} \rightarrow$ standard

6

0.0000008 → sci. notation

8

$2.7 \times 10^{-3} \rightarrow$ standard

REVIEW · B Operations

1

$(3 \times 10^4)(3 \times 10^5)$

3

$(5 \times 10^6)(4 \times 10^3)$

5

$(7 \times 10^5)(8 \times 10^2)$

2

$(8 \times 10^9) \div (2 \times 10^3)$

4

$(9.6 \times 10^8) \div (1.6 \times 10^2)$

6

$(1.2 \times 10^{-3})(4 \times 10^5)$

1. A red blood cell is about 8×10^{-6} m wide; a platelet about 2×10^{-6} m. How many times wider is the red blood cell?

Answer: _____

2. Earth is about 1.5×10^8 km from the Sun. A spacecraft travels 3×10^4 km/h. How many hours for the trip?

Answer: _____

3. A factory makes 5×10^3 widgets per day. How many in 2×10^2 days?

Answer: _____

4. A virus is about 3×10^{-7} m and a cell about 6×10^{-5} m. How many times larger is the cell?

Answer: _____

Unit 2 — Scientific Notation

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

2.2 Scientific Notation. Large → +exp; Small → -exp. Multiply: MULTIPLY coefficients, ADD exponents, adjust to 1-10. Divide: DIVIDE coefficients, SUBTRACT exponents. Examples: $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$; $(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$; $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 \rightarrow 2.0 \times 10^6$.

Practice A — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025 11) 9.04×10^{-7} 12) 7,150,000 13) 6.34×10^{10} 14) 0.0000088

Practice B — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3 5) 3.0×10^{10} 6) 6×10^4 7) 4×10^{11} 8) 2.0×10^4

Practice C — Word Problems

1) $(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ s (500 s) 2) 40× longer 3) $1 \times 10^5 = \$100,000$ 4) 5.2×10^{15} g 5) 3.7×10^8 m 6) 3×10^2 s (300 s) 7) $(2 \times 10^{12}) \div (4 \times 10^6) = 5 \times 10^5$ photos 8) $(3 \times 10^8)(6 \times 10^3) = 1.8 \times 10^{12}$ m

Cumulative Review

A • Conversions: 1) 7.2×10^6 2) 4.5×10^{-4} 3) 30,500,000 4) 0.000061 5) 9.1×10^4 6) 8×10^{-7} 7) 4,440,000 8) 0.0027

B • Operations: 1) 9×10^9 2) 4×10^6 3) 2×10^{10} 4) 6×10^6 5) 5.6×10^8 6) 4.8×10^2

C • Word Problems: 1) 4× wider 2) 5×10^3 h (5,000 h) 3) 1×10^6 widgets 4) $(6 \times 10^{-5}) \div (3 \times 10^{-7}) = 2 \times 10^2 = 200 \times$ larger